BI-ALGEBRAIC p-ADIC HAHN SERIES

by

Yijun Yuan

Let p be a prime number. Let $\mathbf{L}_p := W(\overline{\mathbf{F}}_p)(\!(p^{\mathbf{Q}})\!)$, $\mathbf{L}_p^\flat := \overline{\mathbf{F}}_p(\!(t^{\mathbf{Q}})\!)$, and

$$\Theta\colon \mathbf{L}_p^\flat \longrightarrow \mathbf{L}_p,\ \sum f(s)t^s \longmapsto \sum [f(s)]p^s.$$

Definition 0.1. —

1. *For* $\underline{d} \in \bigoplus_{\mathbf{Z}_{\geq 1}} \mathbf{N}$, *let* $\Sigma(\underline{d}) := \sum_{i=1}^{\infty} d_i$ *and* $\|\underline{d}\| := \sum_{i=1}^{\infty} d_i p^{-i}$.
2. *Let* $\mathbb{P} := \bigoplus_{\mathbf{Z}_{\geq 1}} \{0, 1, \cdots, p-1\} \subsetneq \bigoplus_{\mathbf{Z}_{\geq 1}} \mathbf{N}$. *For any* $\underline{d} := (d_i) \in \mathbb{P}$ *and any integers* $j \geq 1$, *define the series*

$$\underline{d}_j(n) := \sum_{1 \leq i < j} d_i p^{-i} + p^{-n} \sum_{i \geq j} d_i p^{-i},\ n = 0, 1, 2, \cdots.$$

Lemma 0.2. — *For any* $\underline{d} \in \bigoplus_{\mathbf{Z}_{\geq 1}} \mathbf{N}$, *there exists a unique element* $\tau(\underline{d}) \in \mathbb{P}$ *such that* $\|\underline{d}\| - \|\tau(\underline{d})\| \in \mathbf{Z}$.

Proof. — If one writes $\|\underline{d}\|$ as a decimal expansion[1] in base p:

$$\|\underline{d}\| = w.d_1 \cdots d_n \cdots,$$

where $w \in \mathbf{Z}$ and $d_i \in \{0, 1, \cdots, p-1\}$ for any i, then $\tau(\underline{d}) = 0.d_1 \cdots d_n \cdots$. The uniqueness is trivial. $\square$

We collect several properties of these concepts in the following lemma:

lem:18813 ***Lemma 0.3.*** — *Let* $\underline{d}, \underline{e} \in \bigoplus_{\mathbf{Z}_{\geq 1}} \mathbf{N}$.

it:28363 1. *The maps* $\Sigma(\cdot)$ *and* $\|\cdot\|$ *are linear.*
it:52935 2. *One has* $\tau(\underline{d}) = \tau(\underline{e})$ *if and only if* $\|\underline{d}\| - \|\underline{e}\| \in \mathbf{Z}$.
3. $\underline{d} \in \mathbb{P}$ *if and only if* $\tau(\underline{d}) = \underline{d}$.
lem:17816 4. *One has* $\Sigma(\tau(\underline{d})) \leq \Sigma(\underline{d})$. *The equality holds if and only if* $\underline{d} \in \mathbb{P}$.

Proof. — The first three statements are straightforward. We only show the last statement.

For any $\underline{d} \in \bigoplus_{\mathbf{Z}_{\geq 1}} \mathbf{N}$, let

$$N(\underline{d}) := \begin{cases} 0, & \text{if } \underline{d} \in \mathbb{P}; \\ \max\{i \mid d_i \geq p\}, & \text{otherwise.} \end{cases}$$

By a descent argument on $N(\underline{d})$, one only needs to show that if $N(\underline{d}) \geq 1$, then there exists $\underline{e} \in \bigoplus_{\mathbf{Z}_{\geq 1}} \mathbf{N}$ such that $\|\underline{d}\| - \|e\| \in \mathbf{Z}$, $\Sigma(\underline{e}) < \Sigma(\underline{d})$, and $N(\underline{e}) < N(\underline{d})$.

---

[1] We do not allow infinite strings of $(p-1)$ in the decimal expansion of $\|\underline{d}\|$ to ensure the uniqueness of $\tau(\underline{d})$.

If we write $d_{N(\underline{d})} = p\cdot r + s$ with $r \in \mathbf{N}$ and $s \in \{0,1,\cdots,p-1\}$, then one can take $\underline{e} := (d_0,\cdots,d_{N(\underline{d})-2}, d_{N(\underline{d})-1}+r, s, 0, \cdots)$. Then

$$\Sigma(\underline{d}) - \Sigma(\underline{e}) = d_{N(\underline{d})} - r - s = (p-1)\cdot r > 0$$

and $\|\underline{d}\| - \|\underline{e}\| = 0$. The result follows. $\square$

1. A combinatorial condition

def:38144 ***Definition 1.1.*** — *Let p be a prime number. A subset $S \subset \mathbb{P}$ is **sparse** if*

1. *There exists $c \geq 1$ such that $\Sigma(\underline{d}) \leq c$ for all $\underline{d} \in S$;*
2. *For infinitely many $n \geq 1$, there exists n (not necessarily distinct) elements $\underline{d}^{(1)}, \underline{d}^{(2)}, \cdots, \underline{d}^{(n)} \in S$ such that*
   (a) $\Sigma\left(\underline{d}^{(i)}\right) = c$ for $i = 1,2,\cdots,n$ and $\sum_{i=1}^n d_j^{(i)} < p$ for $j \in \mathbf{N}$ (i.e. $\sum_{i=1}^n \underline{d}^{(i)} \in \mathbb{P}$);
   (b) If $\underline{e}^{(1)}, \underline{e}^{(2)}, \cdots, \underline{e}^{(n)} \in S$ satisfy that $\left\|\sum_{i=1}^n \underline{d}^{(i)}\right\| - \left\|\sum_{i=1}^n \underline{e}^{(i)}\right\| \in \mathbf{Z}$, then up to a permutation of $\{1,2,\cdots,n\}$, $\underline{d}^{(i)} = \underline{e}^{(i)}$ for any $i = 1,2,\cdots,n$.

thm:23791 ***Theorem 1.2.*** — *Let $f = \sum_{s\in\mathbf{Q}}[f(s)]t^s \in \mathbf{L}_p^\flat$ be p-adic Hahn series with $\mathrm{Supp}(f) \subseteq \frac{1}{T_f}\mathbf{Z}[p^{-1}]$ for some integer $T_f \geq 1$. If there exists a sparse subset $S \subset \mathbb{P}$ such that $\|S\| := \{\|\underline{d}\| \in \mathbf{Q} | \underline{d} \in S\}$ is a set of representatives of the set $-T_f \cdot \mathrm{Supp}(f)$ modulo $\mathbf{Z}$, then f is transcendental over $\breve{\mathbf{Q}}_p$, and hence transcendental over $\mathbf{Q}_p$.*

Proof. — Let S be a sparse subset of $\mathbb{P}$ in the statement of the theorem and let c be the integer in (1) of Definition 1.1 for S.

One has

$$f = \sum_{\alpha\in\mathrm{Supp}(f)/\frac{1}{T_f}\mathbf{Z}} \sum_{\beta\in\frac{1}{T_f}\mathbf{Z}} [f(\alpha+\beta)]p^{\alpha+\beta} = \sum_{\alpha\in-(T_f\cdot\mathrm{Supp})/\mathbf{Z}} \sum_{\beta\in\mathbf{Z}} \left[f\left(\frac{-\alpha+\beta}{T_f}\right)\right] p^{\frac{-\alpha+\beta}{T_f}}$$
$$= \sum_{\alpha\in\|S\|} C_\alpha p^{-\frac{\alpha}{T_f}} = \sum_{\underline{d}\in S} C_{\|\underline{d}\|} p^{\frac{\|\underline{d}\|}{-T_f}},$$

where

$$C_\alpha := \left(\sum_{\beta\in\mathbf{Z}} \left[f\left(\frac{-\alpha+\beta}{T_f}\right)\right] p^{\frac{\beta}{T_f}}\right) \in W\left(\overline{\mathbf{F}}_p\right)\left(p^{\frac{1}{T_f}}\right)$$

for every $\alpha \in \|S\|$. Notice that $\|S\|$ is a set of representatives of $-T_f\cdot\mathrm{Supp}(f)$ modulo $\mathbf{Z}$, there must exists $\beta \in \mathbf{Z}$ for each $\alpha \in \|S\|$ such that $f\left(\frac{-\alpha+\beta}{T_f}\right) \neq 0$, hence $C_\alpha \neq 0$ for any $\alpha \in \|S\|$.

Suppose that f is a root of a monic polynomial $P(X) = X^n + a_{n-1}X^{n-1} + \cdots + a_0 \in \breve{\mathbf{Q}}_p[X]$. By multiplying $P(X)$ by a suitable power of X, we may assume that there exists n elements $\underline{d}^{(1)}, \underline{d}^{(2)}, \cdots, \underline{d}^{(n)} \in S$ that satisfy the condition (2) of Definition 1.1 for n and c.

The multinomial expansion of $P(f)$ gives

$$0 = \sum_{\substack{\phi:\, S\to\mathbf{N} \\ \sum_{\underline{d}\in S}\phi(\underline{d})\leq n}} C(\phi) p^{-\frac{1}{T_f}\sum_{\underline{d}\in S}\|\underline{d}\|\cdot\phi(\underline{d})},$$

where

$$C(\phi) := a_{\sum_{\underline{d}\in S}\phi(\underline{d})} \cdot \frac{\left(\sum_{\underline{d}\in S}\phi(\underline{d})\right)!}{\prod_{\underline{d}\in S}\phi(d)!} \cdot \left(\prod_{\underline{d}\in S} C_{\|\underline{d}\|}^{\phi(\underline{d})}\right) \in W(\overline{\mathbf{F}}_p)\left(p^{\frac{1}{T_f}}\right).$$

By grouping the terms in the above sum according to the value of $-\frac{1}{T_f}\sum_{\underline{d}\in S}\|\underline{d}\|\cdot\phi(\underline{d})$ modulo $\mathbf{Z}$, one gets

eq:11841

$$\begin{aligned}
0 &= \sum_{r\in\left[0,\frac{1}{T_f}\right)\cap\mathbf{Q}} \sum_{\substack{\phi\colon S\to\mathbf{N}\\ \sum_{\underline{d}\in S}\phi(\underline{d})\le n\\ -\frac{1}{T_f}\sum_{\underline{d}\in S}\|\underline{d}\|\cdot\phi(\underline{d})-r\in\mathbf{Z}}} C(\phi)p^{-\frac{1}{T_f}\sum_{\underline{d}\in S}\|\underline{d}\|\cdot\phi(\underline{d})}\\
&= \sum_{r\in[0,1)\cap\mathbf{Q}} p^{\frac{r}{T_f}} \sum_{\substack{\phi\colon S\to\mathbf{N}\\ \sum_{\underline{d}\in S}\phi(\underline{d})\le n\\ \sum_{\underline{d}\in S}\|\underline{d}\|\cdot\phi(\underline{d})+r\in\mathbf{Z}}} C(\phi)p^{-\frac{1}{T_f}\left(\sum_{\underline{d}\in S}\|\underline{d}\|\cdot\phi(\underline{d})+r\right)},
\end{aligned}\tag{a}$$

where

$$\sum_{\substack{\phi\colon S\to\mathbf{N}\\ \sum_{\underline{d}\in S}\phi(\underline{d})\le n\\ \sum_{\underline{d}\in S}\|\underline{d}\|\cdot\phi(\underline{d})+r\in\mathbf{Z}}} C(\phi)p^{-\frac{1}{T_f}\left(\sum_{\underline{d}\in S}\|\underline{d}\|\cdot\phi(\underline{d})+r\right)} \in W(\overline{\mathbf{F}}_p)\left(p^{\frac{1}{T_f}}\right).$$

By Lemma 1.3, one obtains

eq:39842

$$\sum_{\substack{\phi\colon S\to\mathbf{N}\\ \sum_{\underline{d}\in S}\phi(\underline{d})\le n\\ \sum_{\underline{d}\in S}\|\underline{d}\|\cdot\phi(\underline{d})+r\in\mathbf{Z}}} C(\phi)p^{-\frac{1}{T_f}\left(\sum_{\underline{d}\in S}\|\underline{d}\|\cdot\phi(\underline{d})\right)} = 0 \tag{b}$$

for any $r\in[0,1)\cap\mathbf{Q}$.

Let

$$\phi_0\colon S\longrightarrow\mathbf{N},\ \underline{d}\longmapsto|\{i|\underline{d}=\underline{d}^{(i)}\}|$$

and let $r_0\in[0,1)\cap\mathbf{Q}$ be the unique rational number such that $\sum_{\underline{d}\in S}\|\underline{d}\|\cdot\phi_0(\underline{d})+r_0\in\mathbf{Z}$.

We claim that ϕ_0 is the unique function $\phi\colon S\to\mathbf{N}$ such that $\sum_{\underline{d}\in S}\phi(\underline{d})\le n$ and $\sum_{\underline{d}\in S}\|\underline{d}\|\cdot\phi(\underline{d})+r_0\in\mathbf{Z}$. To show this, suppose that $\phi_1\colon S\to\mathbf{N}$ is another function such that $\sum_{\underline{d}\in S}\phi_1(\underline{d})\le n$ and $\sum_{\underline{d}\in S}\|\underline{d}\|\cdot\phi_1(\underline{d})+r_0\in\mathbf{Z}$.

1. If $\sum_{\underline{d}\in S}\phi_1(\underline{d})<n$, then by Lemma 0.3 (4) we have

$$\Sigma\left(\tau\left(\sum_{\underline{d}\in S}\underline{d}\cdot\phi_1(\underline{d})\right)\right)\le\Sigma\left(\sum_{\underline{d}\in S}\phi_1(\underline{d})\cdot\underline{d}\right)=\sum_{\underline{d}\in S}\phi_1(\underline{d})\sum_{i=1}^{\infty}\underline{d}_i<n\cdot c.$$

On the other hand, the sparse condition (2a) of Definition 1.1 implies that $\sum_{j=1}^{n}d_i^{(j)}<p$ for any i, hence

$$\sum_{\underline{d}\in S}\underline{d}\cdot\phi_0(\underline{d})=\left(\sum_{j=1}^{n}d_i^{(j)}\right)_{i\in\mathbf{Z}_{\ge 0}}=\tau\left(\sum_{\underline{d}\in S}\underline{d}\cdot\phi_0(\underline{d})\right)$$

and consequently

$$\Sigma\left(\tau\left(\sum_{\underline{d}\in S}\underline{d}\cdot\phi_0(\underline{d})\right)\right)=\sum_{i=0}^{\infty}\left(\sum_{j=1}^{n}d_i^{(j)}\right)=\sum_{j=1}^{n}\Sigma(d^{(j)})=n\cdot c.$$

This leads to a contradiction by Lemma 0.3 (2).

2. If $\sum_{\underline{d}\in S}\phi_1(\underline{d})=n$, then we take $\underline{e}^{(1)},\underline{e}^{(2)},\cdots,\underline{e}^{(n)}\in S$ such that $\phi_1(\underline{d})=|\{i|\underline{d}=\underline{e}^{(i)}\}|$ for any $\underline{d}\in S$. Then

$$\sum_{\underline{d}\in S}\|\underline{d}\|\cdot\phi_1(\underline{d})-\sum_{\underline{d}\in S}\|\underline{d}\|\cdot\phi_0(\underline{d})=\left\|\sum_{\underline{d}\in S}\underline{d}\cdot\phi_1(\underline{d})\right\|-\left\|\sum_{\underline{d}\in S}\underline{d}\cdot\phi_0(\underline{d})\right\|=\left\|\sum_{i=1}^{n}\underline{e}^{(i)}\right\|-\left\|\sum_{i=1}^{n}\underline{d}^{(i)}\right\|\in\mathbf{Z}.$$

By the sparse condition (2b) of Definition 1.1, up to a permutation of $\{1,2,\cdots,n\}$, $\underline{d}^{(i)}=\underline{e}^{(i)}$ for any $i=1,2,\cdots,n$. This implies that $\phi_1=\phi_0$.

By the claim, one can specialize (b) at ϕ_0 to get

$$C(\phi_0)=a_{\sum_{\underline{d}\in S}\phi_0(\underline{d})}\cdot\frac{\left(\sum_{\underline{d}\in S}\phi_0(\underline{d})\right)!}{\prod_{\underline{d}\in S}\phi_0(d)!}\cdot\left(\prod_{\underline{d}\in S}C_{\|\underline{d}\|}^{\phi_0(\underline{d})}\right)=0.$$

Since $P(X)$ is monic, $a_{\sum_{\underline{d}\in S}\phi_0(\underline{d})}=1$. Thus, $C_{\|d^{(i)}\|}=0$ for some i, which contradicts the fact that $C_\alpha\neq 0$ for any $\alpha\in\|S\|$. Hence f is transcendental over $\breve{\mathbf{Q}}_p$.

□

lem:35613 ***Lemma 1.3.*** — *Let $T\geq 1$ be an integer. Every p-adic Hahn series can be uniquely written as $\sum_{s\in[0,T^{-1})}c_s\cdot p^s$, with $c_s\in\breve{\mathbf{Q}}_p\left(p^{\frac{1}{T}}\right)$ for all s.*

Proof. — Notice that $\mathbf{Q}\cap\left[0,\frac{1}{T}\right)$ is a set of representatives of $\mathbf{Q}/\frac{1}{T}\mathbf{Z}$, every p-adic Hahn series $\sum_{q\in\mathbf{Q}}[a(q)]p^q$ can be written as

$$\sum_{q\in\mathbf{Q}}[a(q)]p^q=\sum_{q\in\mathbf{Q}\cap\left[0,\frac{1}{T}\right)}\sum_{n\in\frac{1}{T}\mathbf{Z}}[a(q+n)]p^{q+n}=\sum_{q\in\mathbf{Q}\cap\left[0,\frac{1}{T}\right)}p^q\left(\sum_{n\in\frac{1}{T}\mathbf{Z}}[a(q+n)]p^n\right),$$

where $\sum_{n\in\frac{1}{T}\mathbf{Z}}[a(q+n)]p^n\in\breve{\mathbf{Q}}_p\left(p^{\frac{1}{T}}\right)$ for all q.

For the uniqueness, for any p-adic Hahn series $\sum_{q\in\mathbf{Q}}[a(q)]p^q$, write

$$\sum_{q\in\mathbf{Q}}[a(q)]p^q=\sum_{q\in\mathbf{Q}\cap\left[0,\frac{1}{T}\right)}c_q\cdot p^q=\sum_{q\in\mathbf{Q}\cap\left[0,\frac{1}{T}\right)}d_q\cdot p^q$$

with $c_q,d_q\in\breve{\mathbf{Q}}_p\left(p^{\frac{1}{T}}\right)$ for all $q\in\mathbf{Q}\cap\left[0,\frac{1}{T}\right)$. Then we have

$$0=\sum_{q\in\mathbf{Q}\cap\left[0,\frac{1}{T}\right)}(c_q-d_q)p^q.$$

If we write $c_q-d_q=\sum_{n\in\frac{1}{T}\mathbf{Z}}[s_{q,n}]p^n\in\breve{\mathbf{Q}}_p\left(p^{\frac{1}{T}}\right)$ with $s_{q,n}\in\overline{\mathbf{F}}_p$ for all n and for all $q\in\mathbf{Q}\cap\left[0,\frac{1}{T}\right)$, then

eq:19959 (c)
$$0=\sum_{q\in\mathbf{Q}\cap\left[0,\frac{1}{T}\right)}\sum_{n\in\frac{1}{T}\mathbf{Z}}[s_{q,n}]p^{q+n}.$$

Since $q+n$ are all distinct for different pairs (q,n), (c) is the standard expansion of $0\in\mathscr{O}_{\breve{\mathbf{Q}}_p}(\!(p^{\mathbf{Q}})\!)$. Hence $s_{q,n}=0$ for all $q\in\mathbf{Q}\cap\left[0,\frac{1}{T}\right)$ and $n\in\frac{1}{T}\mathbf{Z}$, i.e. $c_q=d_q$ for all $q\in\mathbf{Q}\cap\left[0,\frac{1}{T}\right)$. □

Yijun Yuan, Institute for Theoretical Sciences, Westlake University, No. 600 Dunyu Road, Sandun town, Xihu district, Hangzhou, Zhejiang Province, 310030, China • *E-mail :* 941201yuan@gmail.com
Url : https://yijunyuan.github.io/